the end of the timestep, and (iii) there is a single collision within the timestep. [The case of additional collisions within the step could be handled as in (iii).] In cases (i) and (ii) it is clear that the error accumulating in a single step will be third order in the stepsize, since the Verlet method has local error of order three (since it is a second order method).

Now consider case (iii). The contact happens when the particles are separated by a distance 1, at which point $\boldsymbol{u}_\perp = \boldsymbol{q}$. We assume the time of contact is τ_c.

$$\bar{\boldsymbol{P}} := \boldsymbol{p} - \tfrac{h}{2}\nabla U(\boldsymbol{q}), \quad \boldsymbol{q}_c := \boldsymbol{q} + \tau_c \bar{\boldsymbol{P}},$$

$$\boldsymbol{u}_\perp := \boldsymbol{q}_c, \qquad \hat{\boldsymbol{P}} := \bar{\boldsymbol{P}} + \alpha\, \boldsymbol{u}_\perp,$$

$$\boldsymbol{Q} := \boldsymbol{q}_c + (h - \tau_c)\hat{\boldsymbol{P}} \quad \boldsymbol{P} := \hat{\boldsymbol{P}} - \tfrac{h}{2}\nabla U(\boldsymbol{Q}),$$

The coefficient α is chosen so that the kinetic energy is conserved through collision, thus

$$\alpha = -2\frac{\boldsymbol{u}_\perp \cdot \bar{\boldsymbol{P}}}{\boldsymbol{u}_\perp \cdot \boldsymbol{u}_\perp}.$$

We next calculate the change in energy in a single step by inserting $\boldsymbol{Q}$ and $\boldsymbol{P}$ into the Hamiltonian and expanding around the point of collision $\boldsymbol{q}_c$, obtaining

$$\begin{aligned}\Delta H &\overset{\text{def}}{=} H(\boldsymbol{Q},\boldsymbol{P}) - H(\boldsymbol{q},\boldsymbol{p}) \\ &= -(h - 2\tau_c)\frac{\boldsymbol{q}_c^T \bar{\boldsymbol{P}}}{\boldsymbol{q}_c^T \boldsymbol{q}_c}\boldsymbol{q}_c^T \nabla U(\boldsymbol{q}_c) + O\left(h^2\right)\end{aligned}$$

Energy is therefore conserved only to first order through the collision unless one of the following conditions holds:

- there is no collision
- the collision occurs at the end of a timestep
- the collision occurs at the middle of the timestep, $\tau_c = h/2$,
- the directional derivative of U along the collision vector ($\boldsymbol{u}_\perp = \boldsymbol{q}_c$) vanishes, or
- the momentum vector is orthogonal to the collision vector at the point of contact.

In general, because an error of size $\mathcal{O}(h)$ occurs in each collision and there are a finite number of collisions in a fixed time interval, the total error is also $\mathcal{O}(h)$, i.e. first order.